

extra strength, at the expense of increasing drag. Platz stuck to wood as his prime material, although he used steel tubing for the internal structure of the fuselage and adopted a high wing in contrast to the low wing preferred by Junkers. Fokker monoplanes were much admired in the 1920s, especially the F.VII-3m (trimotor), flown by aviators such as Kingsford Smith and Byrd. However, both wooden construction and a high wing were to prove retrograde. When retractable undercarriages became common in the 1930s, high-wing aircraft had nowhere to retract the landing gear into. The Junkers corrugated metal designs also had limitations. By 1920 another German designer, Dr. Adolph Rohrbach, had worked out the advantages of making wings and tail surfaces out of a smooth metal skin stretched over box spars. This “stressed skin” would bear some of the load previously borne entirely by the frame. Rohrbach’s ideas for a smooth metal airplane did not come to the attention of American aircraft designers until 1926 but were to prove immensely influential on Jack Northrop and Boeing, and then on almost all manufacturers.

Reducing drag

Progress in the theory of aerodynamics, backed up by research at well-funded laboratories such as America’s National Advisory Committee for

Aeronautics (NACA) facility in Virginia, also began to have a profound effect from the second half of the 1920s. NACA’s array of wind tunnels was especially important in studying airflow around different models of airplane. Some of its experiments fed directly into aircraft design. For example, air-cooled radial engines were being widely adopted in the 1920s because of their excellent weight-to-horsepower ratios, but their exposed ring of engine cylinders tended to have an adverse effect on streamlining. In 1928 NACA wind-tunnel experiments with various forms of cowling showed that a full cowling of the right design could eliminate 60 percent of the drag from the engine while actually improving cooling. The “NACA cowling” became a standard feature of airplanes with radial engines, producing a significant increase in performance.

The Lockheed Vega, designed by Jack Northrop, was one of the first airplanes to adopt the NACA cowling in the late 1920s. With its Fokker-influenced high, strut-free single wing and streamlined monocoque fuselage, the Vega represented one of the most advanced designs of its day – not surprisingly embraced by the likes of Post and Earhart. The fact that its fuselage was made of two easily assembled prefabricated parts also looked forward to the increasing use of mass-production methods in the air industry in place of

traditional craft skills. But the Vega was made of spruce wood and had a fixed undercarriage, two features that were soon to make it seem dated.

Established innovations

The 1930s brought to fruition the revolution in aircraft design begun in the previous decade. Monoplanes completed their triumph over high-drag biplanes, while all-metal stressed-skin construction became the rule, benefiting from improved metallurgy, especially lightweight aluminum alloys (aircraft manufacture was the first major use that had been found for aluminum). Engines continued to improve in power-to-weight ratio and reliability. Two types prevailed: air-cooled radials such as the Pratt & Whitney Wasp series, especially favored in the United States; and liquid-cooled in-line engines such as the Rolls-Royce Merlin. The power of radial engines was increased by adding a second ring of cylinders, and the NACA cowling made them more efficient. In-line engines were

DAREDEVIL DOOLITTLE

James Doolittle, wearing a flying suit and parachute, refuels the center wing tank of his Laird Super Solution racing plane.

Doolittle went down in history for making the first blind flight.

He flew a 15-mile (24-km) irregular course in a Consolidated NT-2 biplane before landing safely.

